

Can Raising the 15m Score Floor Close the Gap to 30%?

One lever, isolated, against the same 6 fixed windows as the Baseline Campaign — 15m only, 1h untouched

Generated 2026-06-21 · BTC/ETH/SOL · floor 50 (baseline) vs 60 vs 65

The Baseline Campaign found 1h near-target (+26.9% mean, 89% of sub-runs profitable) while 15m dragged the combined average down (+14.2% mean, only 50% profitable, PF 1.07 — barely above breakeven). Its own verdict identified this as an **entry-quality** problem on 15m, not a sizing problem. This batch tests the most direct lever for that: raising the score floor (the final post-adjustment threshold below which a signal is rejected outright) from 50 to 60 to 65, 15m only, against the exact same 6 historical windows used in the Baseline Campaign, so the comparison is clean. 1h is intentionally untouched.

A Real Constraint Found Before Running Anything

The brief assumed the score floor could be set via the existing `simConfigOverrides` passthrough without touching `simulator.ts`. That assumption didn't hold: **the score floor was a hardcoded module constant (`const SCORE_FLOOR = 50`), not a `SimConfig` field** — there was no config knob for `simConfigOverrides` to reach. Running this batch at all required one minimal, additive change: `scoreFloor?: number` was added to `SimConfig`, defaulting to 50 everywhere (`defaultSimConfig()` now sets it explicitly, and the one usage site reads `config.scoreFloor ?? 50`) — every existing caller that doesn't set it keeps today's exact behavior, unchanged. No scoring, sizing, or guardrail logic was touched; an existing magic number was parameterized, the same pattern `minRR / minConfidence` already follow. A matching slider was added to the `/simulate` page so the value could actually be set and tested.

Secondary Change — Guardrail Counter Split

As requested, the previously-shared `tradesRejectedByDailyLimit` counter (incremented by both the daily 3% and weekly 6% loss-limit checks) was split into separate `tradesRejectedByDailyLimit` and `tradesRejectedByWeeklyLimit` counters. This was a clean, unambiguous split — the code already had separate `if` blocks for the daily and weekly checks; the weekly block was simply incrementing the daily counter by a naming oversight. No detection logic changed, only which counter a rejection increments.

Condition A (floor 50) uses the Baseline Campaign's already-recorded data, predating this split. Per the brief's own instruction, that data is reported with the daily/weekly counts combined (marked "n/a*" for weekly in the appendix) rather than fabricating a retroactive split that the saved data doesn't actually contain. Conditions B and C, run after the split, have full daily/weekly detail.

Condition Comparison — The Core Result

CONDITION	MEAN RETURN	MEAN PF	MEAN WR	PROFITABLE SUB-RUNS	MEAN MAX DD	MEAN TRADES
Floor 50 (baseline)	+14.2%	1.07	48.9%	9/18 (50%)	20.4%	186.2
Floor 60	+20.4%	1.12	51.0%	10/18 (56%)	19.5%	173.4
Floor 65	+20.4%	1.12	51.8%	11/18 (61%)	19.3%	168.1

Raising the floor from 50 to 60 produced a real improvement — mean return +14.2%→+20.4%, PF 1.07→1.12, win rate 48.9%→51.0%, profitable sub-runs 50%→56%. **Raising it further to 65 did not add anything meaningful on top of 60** — return and PF are statistically identical (20.4%/1.12 both), win rate ticked up slightly (51.8% vs 51.0%) and one more sub-run turned profitable (11 vs 10), but trade volume kept dropping (-9.7% vs -6.9% from baseline) for no corresponding net gain. **This is the diminishing-returns pattern the brief asked to watch for, and it shows up clearly:** floor 60 captures essentially all of the available benefit from this lever; floor 65 just trades away more volume to get the same result.

Trade Volume Impact

CONDITION	MEAN TRADES / WINDOW	CHANGE VS FLOOR 50
Floor 50	186.2	—
Floor 60	173.4	-6.9%
Floor 65	168.1	-9.7%

This is "better PF with a modest trade-count reduction," not "better PF with a third as many trades" — the floor-60 improvement comes from cutting roughly 1 in 14 trades, not from gutting volume. That's a meaningfully different (and more attractive) finding than an aggressive cut would have been: the strategy isn't becoming reliant on a thin slice of trades to look good.

Per-Window Detail

WINDOW	REGIME	FLOOR 50 RETURN	FLOOR 60 RETURN	FLOOR 65 RETURN	BEST CONDITION
1 (2020-10→2021-03)	Bull	+17.5%	+21.5%	+26.1%	65
2 (2023-10→2024-03)	Bull	+14.9%	+27.3%	+29.4%	65
3 (2021-05→2021-10)	Bear	+20.9%	+22.1%	+22.6%	65 (marginal)
4 (2022-04→2022-09)	Bear	+19.0%	+31.3%	+27.0%	60
5 (2023-07→2023-12)	Choppy	-16.3%	-14.9%	-14.8%	65 (marginal, still negative)
6 (2024-04→2024-09)	Choppy (sub.)	+29.1%	+35.2%	+31.8%	60

Window 4 is a direct counterexample to "higher floor is always better": floor 60 (+31.3%) clearly beats floor 65 (+27.0%) there — exactly the kind of result the brief asked to be reported plainly rather than smoothed over. **Window 5 (choppy) stays negative in all three conditions**, barely moving (-16.3% → -14.9% → -14.8%) — score floor is not the lever that fixes this specific window; whatever is wrong with the strategy's behavior in that period is not primarily an entry-quality problem.

Combined 15m + 1h Projected Impact

This is the number that actually matters for the original 30% target — using the Baseline Campaign's unchanged 1h results (+26.9% mean, untouched by this batch) alongside each 15m score-floor condition:

CONFIGURATION	COMBINED MEAN RETURN	WINDOWS ≥30%
Floor 50 + 1h (Baseline Campaign actual)	+20.5%	1/6
Floor 60 + 1h	+23.6%	2/6
Floor 65 + 1h	+23.6%	2/6

Floor 60 (or 65 — they land in the same place combined) moves the blended mean from +20.5% to +23.6%, and pushes one additional window (Window 2, Bull Oct 2023-Mar 2024) over the 30% line. That closes roughly **a third of the gap** to 30% (3.1 of the needed 9.5 percentage points) — a real, worthwhile improvement, but not remotely sufficient on its own.

Guardrail Split Findings

CONDITION	TOTAL DAILY-LIMIT REJECTIONS	TOTAL WEEKLY-LIMIT REJECTIONS	DAILY : WEEKLY RATIO
Floor 60	2,506	6,144	1 : 2.45
Floor 65	2,607	6,368	1 : 2.44

The weekly loss limit (6%) is doing roughly 2.4x more of the rejecting than the daily limit (3%) on 15m — and that ratio is essentially unchanged between floor 60 and floor 65 (2.45 vs 2.44), despite floor 65 letting through fewer signals overall. This answers the brief's open question directly: **raising the score floor did not shift which guardrail does the rejecting**. The weekly limit's dominance looks structural — a function of 15m's trade frequency and loss clustering over a week, not of how strict the entry-quality gate is. Tightening entry quality further would not be expected to meaningfully change this balance.

Explicit Verdict

Raising the 15m score floor from 50 to 60 meaningfully improved PF and win rate, moving from 1.07/48.9% to 1.12/51.0%. Trade volume dropped by 6.9%. **Floor 65 showed diminishing returns relative to floor 60** — statistically identical return and PF, for an additional 2.8-point volume cut and a window (#4) where it actually underperformed floor 60. **Combined with the unchanged 1h results, the blended 15m+1h mean return across the same 6 windows moves from +20.5% to +23.6%** at floor 60, closing roughly a third of the gap to the 30% target — not closing it.

Recommendation: adopt floor 60 for 15m, not 65 — it captures essentially all of this lever's available benefit while keeping more trades (and more self-learning sample volume) than floor 65 for no measurable cost. **This lever alone is insufficient — proceed to exit-logic testing (trailing stop / partial profit-taking) on 15m next**, using floor 60 as the new 15m baseline going into that test, same 6 windows, same discipline. Window 5 (choppy, Jul-Dec 2023) is a specific area to watch in that next test — score floor barely moved it, suggesting its problem is in how trades are managed once open, not which ones are taken.

Appendix — All 54 Sub-Run Results (3 Conditions × 18 Runs)

WINDOW	SYMBOL	SCORE FLOOR	TRADES	WR	PF	RETURN	MAX DD	DAILY REJ.	WEEKLY REJ.
1	BTCUSDT	50	172	48.8%	1.08	+10.3%	17.9%	826	n/a*
1	ETHUSDT	50	179	54.2%	0.97	-3.3%	13.1%	565	n/a*
1	SOLUSDT	50	146	56.8%	1.36	+45.4%	10.2%	373	n/a*
1	BTCUSDT	60	164	52.4%	1.15	+18.5%	16.6%	107	476
1	ETHUSDT	60	177	55.4%	1.07	+8.0%	12.2%	120	180
1	SOLUSDT	60	137	56.2%	1.31	+38.0%	12.3%	157	546
1	BTCUSDT	65	160	54.4%	1.19	+23.8%	15.1%	124	431
1	ETHUSDT	65	175	56.0%	1.12	+14.6%	9.6%	120	180
1	SOLUSDT	65	136	56.6%	1.32	+39.8%	12.3%	157	546
2	BTCUSDT	50	220	45.0%	0.79	-23.9%	24.6%	325	n/a*
2	ETHUSDT	50	181	47.5%	0.95	-6.2%	25.7%	662	n/a*
2	SOLUSDT	50	199	53.3%	1.46	+74.9%	16.8%	82	n/a*
2	BTCUSDT	60	191	46.6%	0.77	-23.7%	24.6%	313	337
2	ETHUSDT	60	171	49.1%	0.98	-2.0%	23.6%	45	617
2	SOLUSDT	60	186	57.0%	1.68	+107.6%	15.6%	144	0
2	BTCUSDT	65	184	47.8%	0.80	-22.5%	23.4%	382	342
2	ETHUSDT	65	164	50.6%	1.02	+2.4%	19.3%	19	677
2	SOLUSDT	65	193	57.5%	1.62	+108.3%	18.9%	161	0
3	BTCUSDT	50	204	45.1%	0.78	-23.8%	25.8%	301	n/a*
3	ETHUSDT	50	169	53.3%	1.31	+49.7%	27.4%	205	n/a*
3	SOLUSDT	50	154	61.7%	1.32	+36.7%	12.4%	357	n/a*

3	BTCUSDT	60	181	47.0%	0.82	-20.7%	28.4%	155	388
3	ETHUSDT	60	153	56.2%	1.34	+50.8%	24.6%	85	140
3	SOLUSDT	60	149	61.7%	1.33	+36.2%	14.1%	94	276
3	BTCUSDT	65	168	48.8%	0.85	-18.8%	29.9%	250	388
3	ETHUSDT	65	151	57.6%	1.38	+58.4%	25.4%	84	140
3	SOLUSDT	65	150	60.7%	1.24	+28.1%	12.3%	102	276
4	BTCUSDT	50	198	40.9%	0.86	-16.7%	26.0%	553	n/a*
4	ETHUSDT	50	193	51.8%	1.32	+43.5%	20.4%	859	n/a*
4	SOLUSDT	50	206	51.5%	1.16	+30.1%	16.7%	346	n/a*
4	BTCUSDT	60	177	45.8%	0.95	-6.7%	23.6%	110	365
4	ETHUSDT	60	186	54.8%	1.39	+55.6%	17.5%	193	283
4	SOLUSDT	60	201	53.2%	1.23	+45.1%	14.9%	107	221
4	BTCUSDT	65	167	47.9%	0.97	-3.9%	23.6%	134	389
4	ETHUSDT	65	173	55.5%	1.34	+44.2%	17.0%	231	463
4	SOLUSDT	65	193	54.4%	1.20	+40.8%	15.9%	114	408
5	BTCUSDT	50	193	42.5%	0.94	-8.8%	23.2%	441	n/a*
5	ETHUSDT	50	153	41.2%	0.78	-21.3%	26.0%	544	n/a*
5	SOLUSDT	50	200	42.5%	0.86	-18.8%	23.5%	389	n/a*
5	BTCUSDT	60	172	43.0%	0.93	-9.1%	23.3%	198	240
5	ETHUSDT	60	150	41.3%	0.79	-21.5%	24.7%	42	324
5	SOLUSDT	60	188	44.1%	0.89	-14.2%	23.3%	270	338
5	BTCUSDT	65	167	46.1%	0.99	-1.2%	18.4%	211	240
5	ETHUSDT	65	144	42.4%	0.78	-21.2%	25.9%	28	238
5	SOLUSDT	65	180	44.4%	0.81	-22.0%	24.3%	145	272
6	BTCUSDT	50	192	47.9%	1.05	+6.2%	15.2%	457	n/a*
6	ETHUSDT	50	190	40.0%	0.61	-31.6%	32.1%	470	n/a*
6	SOLUSDT	50	203	56.9%	1.71	+112.6%	10.0%	627	n/a*
6	BTCUSDT	60	174	54.6%	1.21	+26.7%	12.9%	97	300
6	ETHUSDT	60	170	41.8%	0.69	-29.0%	29.5%	141	520
6	SOLUSDT	60	194	58.0%	1.66	+108.0%	10.0%	128	593
6	BTCUSDT	65	171	52.0%	1.10	+12.3%	16.0%	88	258
6	ETHUSDT	65	157	42.0%	0.70	-27.8%	28.3%	141	520
6	SOLUSDT	65	193	56.8%	1.68	+110.9%	11.0%	116	600

* Condition A (floor 50) predates the daily/weekly counter split — these rows show the combined count under "Daily Rej." with "Weekly Rej." marked n/a, per the brief's instruction not to fabricate a retroactive split.

Report generated from 18 newly-run live simulations per score-floor condition (36 total) plus the Baseline Campaign's existing 18 floor-50 runs, against real historical Binance market data. Zero console errors across both new conditions. Raw data: `training-exports/phase3-floor60-results.json` , `training-exports/phase3-floor65-results.json` , `training-exports/baseline-campaign-results.json` .